Geometry Unit 11 Assessment Guide

General Information	In unit 11, students apply the concepts learned in unit 10 to solve problems involving triangles and quadrilaterals inscribed in a circle (MA.912.GR.6.3). The content also extends to solving problems involving triangles using measures of interior angles of a triangle sum to 180°; measures of a set of exterior angles of a triangle sum to 360°; triangle inequality theorem; base angles of isosceles triangles are congruent; the segment joining midpoints of two sides of a triangle is parallel to the third side and half the length; and the medians of a triangle meet at a point. Students have used or learned several of these relationships in their prior learning or earlier in Geometry. To assess their understanding, students will apply their understanding to setting up and solving equations. The equations that students solve match MA.7.AR.2.2 or MA.8.AR.2.1. If students are struggling solving equations as found in unit 11 and on the unit assessment, teachers are encouraged to support students using grade 7 unit 10 and/or grade 8 unit 8. Question 4 shows students two different incomplete constructions. These constructions are sufficient for the student to know the location of the point of concurrency. The assessment limits for benchmark MA.912.GR.5.3 state: • Items will assess constructions through the use of compass and straight-edge with descriptions, images, and/or animations. • Items may ask the student to describe steps in a construction, identify the next step in a given construction, or describe what construction is being created. In question 4, students use their knowledge of constructions of inscribed and circumscribed circles of a triangle to complete statements. The last statement connects the construction to the measures of interior angles of a triangle sum to $180°$. Question 4 helps teachers understand how constructions could be assessed when students do not have access to geometric tools to perform constructions.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.912.GR.6.3
$n =$ 12		**Recommended Scoring** **(7 Total Points)**	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of n.
Question	**1b**	**Benchmark(s)**	MA.912.GR.6.3
$m\overset{\frown}{PY} =$ 78°		**Recommended Scoring** **(6 Total Points)**	Consider giving students full credit if a student makes an error in 1a but correctly finds $m\overset{\frown}{PY}$.
Question	**2**	**Benchmark(s)**	MA.912.GR.1.3
Least | | **Greatest** $\angle L$ | $\angle S$ | $\angle T$		**Recommended Scoring** **(6 Total Points)**	No partial credit is suggested.

Question	3a	Benchmark(s)	MA.912.GR.1.3
$RC =$ 35.9		**Recommended Scoring** **(7 Total Points)**	Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer.
Question	**3b**	**Benchmark(s)**	MA.912.GR.1.3
Perimeter 71.3		**Recommended Scoring** **(7 Total Points)**	Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer.

Question	**4**	**Benchmark(s)**	MA.912.GR.5.3
Complete the statements. [○ Nate's / ○ Whitney's] construction uses [○ medians / ○ angle bisectors / ○ segment bisectors / ○ perpendicular bisectors] to construct the incenter of ΔHBX. When completed, [○ Nate's / ○ Whitney's] construction will result in a circle that is circumscribed about ΔHBX with [○ $\overline{BN}$ / ○ $\overline{BW}$ / ○ $\overline{LN}$ / ○ $\overline{LW}$] as a radius. [○ Nate's / ○ Whitney's] construction can be used to show the Triangle Sum theorem because the angles of ΔHBX are [○ central angles / ○ inscribed angles / ○ angles formed by tangents] of the [○ inscribed / ○ circumscribed] circle such that the three arcs intercepted by the interior angles of ΔHBX complete a circle, so the sum of the measures of the interior angles of ΔHBX is half the measure of a circle.		**Recommended Scoring** **(14 Total Points)**	Consider giving students points as follows: • Correctly completes the first statement – 4 points. • Correctly completes the second statement – 4 points. • Correctly completes the third statement – 6 points. It is not suggested that students receive partial credit by further dividing up the points.

Question	5a	Benchmark(s)	MA.912.GR.6.3
$x =$ 17		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of x.
Question	**5b**	**Benchmark(s)**	MA.912.GR.6.3
$y =$ 5		**Recommended Scoring** **(7 Total Points)**	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of y.

Question	6	Benchmark(s)	MA.912.GR.1.3
<table><tr><td>$d =$</td><td>17.1</td></tr></table>		**Recommended Scoring** **(7 Total Points)**	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of d.
Question	**7**	**Benchmark(s)**	MA.912.GR.1.3
☐ 14, 14, 14 ☐ 17, 18, 35 ☐ $\frac{37}{4}, \frac{37}{8}, \frac{37}{12}$ ☐ 23.1, 45.6, 73.4 ☐ 45.62, 65.76, 111.21		**Recommended Scoring** **(6 Total Points)**	Consider giving students 3 points for each correct answer. If students have any incorrect choices, then consider deducting 2 points for each incorrect answer.

Question	8a,b	Benchmark(s)	MA.912.GR.1.3
		Recommended Scoring (14 Total Points)	Consider giving students 7 points for each length. Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer.

Segment	Length
$\overline{RW}$	20.125
$\overline{RH}$	24.74

Question	9	Benchmark(s)	MA.912.GR.1.3
		Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correct answer. If students have any incorrect choices, then consider deducting 2 points for each incorrect answer.

- ☐ $\overline{YH} \cong \overline{HB}$
- ☐ $\overline{YR} \cong \overline{BW}$
- ☐ $\Delta RHW \sim \Delta YHB$
- ☐ $\angle HRW \cong \angle HWR$
- ☐ $\angle HWR \cong \angle WBY$

Question	10	Benchmark(s)	MA.912.GR.6.3
		Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correct answer. If students have any incorrect choices, then consider deducting 2 points for each incorrect answer.

- ☐ $m\angle LXC = 90°$
- ☐ $m\angle CNH = 90°$
- ☐ $m\angle XNY + m\angle YLX = 180°$
- ☐ $m\angle MWC + m\angle MHC = 180°$
- ☐ $m\angle MWC + m\angle WMH = 180°$